

ACTIVE PASS FILTER

Aim

To design and set up the active First order filters using Op-amp.

Apparatus Required

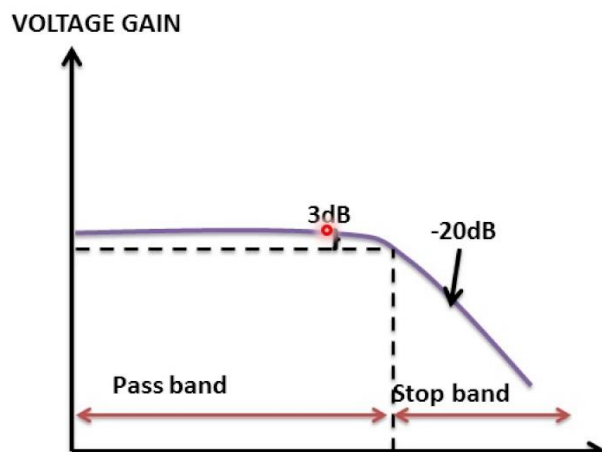
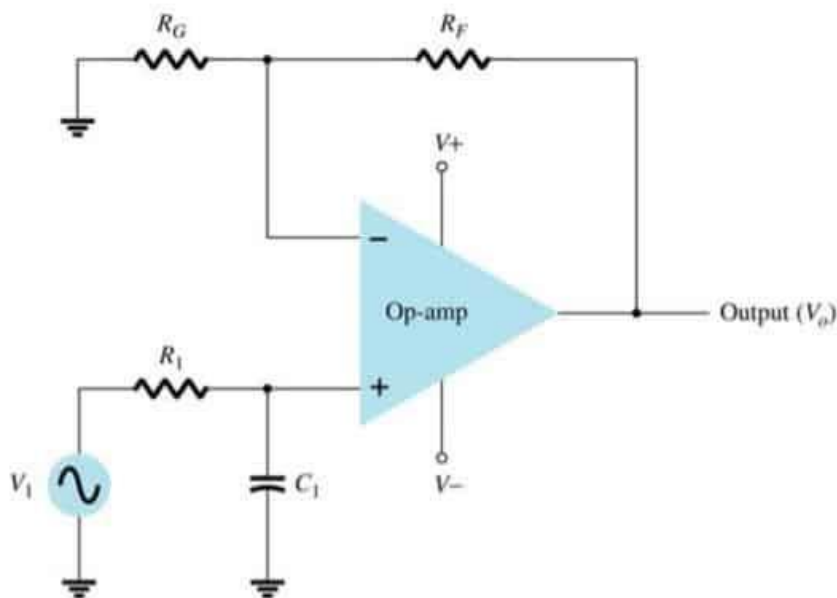
Op-amp, resistors, capacitors, breadboard, CRO, function generator and power supplies.

Design:

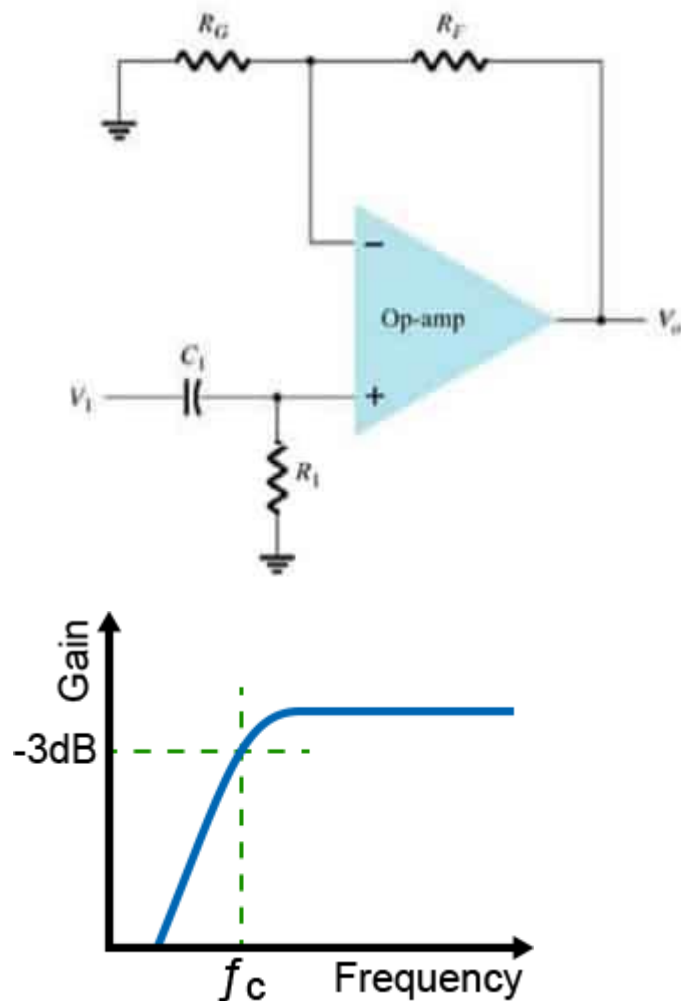
$$F_c = \frac{1}{2\pi RC}$$

Circuit Diagram

First Order LPF



First Order HPF



Procedure

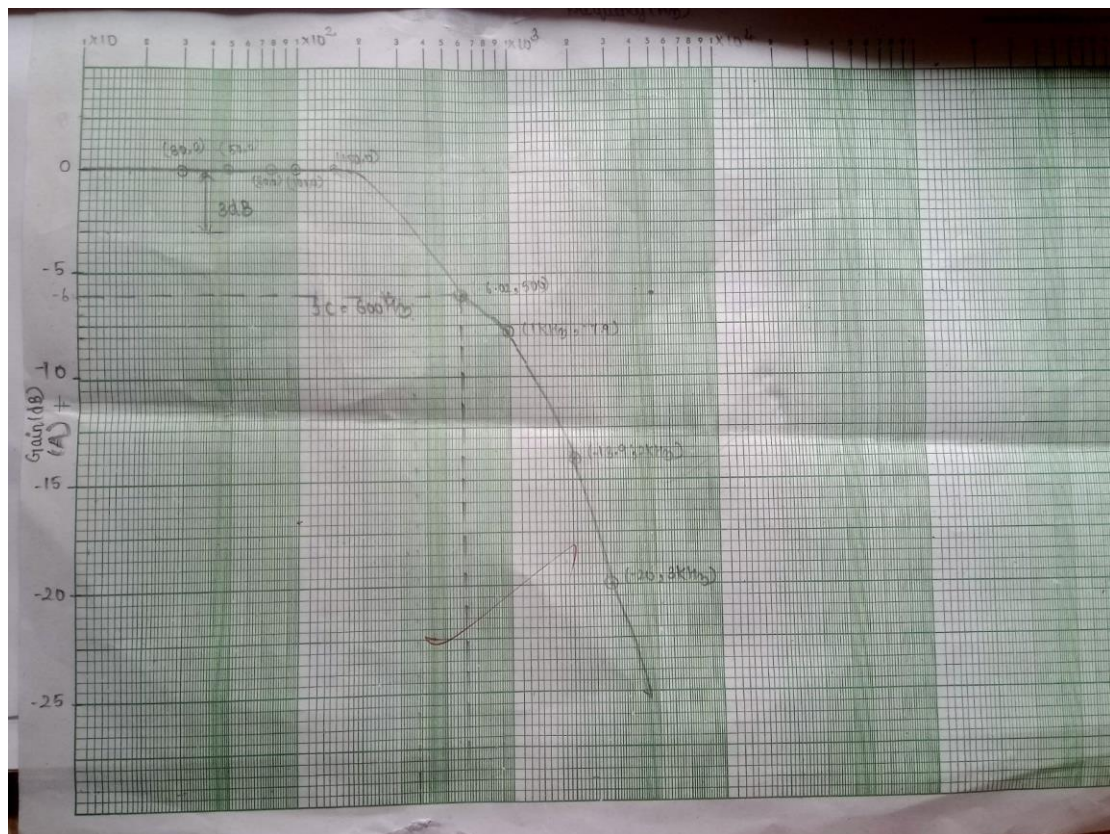
1. Assemble the first-order active filter circuit on the breadboard.
2. Connect IC 741 to a $\pm 15\text{ V DC}$ supply.
3. Connect the function generator to the input of the filter.
4. Apply a sinusoidal input, for example, 1 V Vm
5. Initially set the input frequency well below the calculated cut-off frequency.
6. Observe V_{in} and V_{out} on the CRO/DSO.
7. Measure the output voltage at different input frequencies.
8. Calculate the voltage gain IN Decibel.
9. Increase the frequency gradually through and beyond the expected cut-off frequency.
10. Record the output voltage at each frequency.
11. Identify the frequency at which the gain becomes approximately **-3 dB from the passband gain.**
12. Plot the frequency-response curve of **gain (dB) versus frequency.**

13. Compare the experimentally obtained cut-off frequency with the theoretical value.

Observation Table

S. No.	Frequency (Hz)	Vin (V)	Vout(V)	Gain	Gain (dB)
1	30Hz	1V	1V	1	0
2	50Hz	1V	1V	1	0
3	80Hz	1V	1V	1	0
4	100Hz	1V	1V	1	0
5	150Hz	1V	1V	1	0
6	500Hz	1V	500mV	500	-6.02
7	1KHz	1V	400mV	400	-7.9
8	2KHz	1V	200mV	200	-13.9
9	3KHz	1V	100mV	100	-20

GRAPH:-



Result:

Thus Designed the First order filters using op amp and obtained the results as Cutoff frequency of low pass filter is600Hz.....

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